Section A

Qn	Suggested Solution
1	$y=-x^2+3x-1 \Rightarrow x^2-3x+1+y=0$ $\Rightarrow x=\dfrac{-(-3)\pm\sqrt{(-3)^2-4(1)(1+y)}}{2(1)}$ $\Rightarrow x=\dfrac{3\pm\sqrt{5-4y}}{2}$ Since $x<\dfrac{3}{2}$, $x=\dfrac{3-\sqrt{5-4y}}{2}$ Required volume = $\pi\left[\int_{-1}^{1}\left(\dfrac{3-\sqrt{5-4y}}{2}\right)^2 \mathrm{d}y-\int_0^1\left(y^2\right)^2 \mathrm{d}y\right]$ $=\pi\left[\dfrac{1}{4}\int_{-1}^{1}9-6\sqrt{5-4y}+(5-4y)\,\mathrm{d}y-\int_0^1 y^4\,\mathrm{d}y\right]$ $=\pi\left[\dfrac{1}{2}\int_{-1}^{1}7-2y-3\left(5-4y\right)^{\frac{1}{2}}\,\mathrm{d}y-\int_0^1 y^4\,\mathrm{d}y\right]$ $=\dfrac{\pi}{2}\left[7y-y^2+\dfrac{1}{2}\left(5-4y\right)^{\frac{3}{2}}\right]_{-1}^{1}-\left[\dfrac{\pi}{5}y^5\right]_0^1$ $=\dfrac{\pi}{2}\left[\left(7-1+\dfrac{1}{2}\right)-\left(-7-1+\dfrac{1}{2}9^{\frac{3}{2}}\right)\right]-\dfrac{\pi}{5}$ $=\dfrac{\pi}{2}\left(\dfrac{13}{2}-\dfrac{11}{2}\right)-\dfrac{\pi}{5}=\dfrac{\pi}{2}-\dfrac{\pi}{5}=0.3\pi$ units^3.

2(a)(i)	$x=k\cos t \Rightarrow \dfrac{\mathrm{d}x}{\mathrm{d}t}=-k\sin t$ $y=k\sin 2t \Rightarrow \dfrac{\mathrm{d}y}{\mathrm{d}t}=2k\cos 2t$ $\therefore \dfrac{\mathrm{d}y}{\mathrm{d}x}=\dfrac{\mathrm{d}y}{\mathrm{d}t}\dfrac{\mathrm{d}t}{\mathrm{d}x}=\dfrac{2k\cos 2t}{-k\sin t}=-\dfrac{2\cos 2t}{\sin t}$. At $(k\cos p, k\sin 2p)$, $t=p \Rightarrow \dfrac{\mathrm{d}y}{\mathrm{d}x}=-\dfrac{2\cos 2p}{\sin p}$. Gradient of normal = $\dfrac{\sin p}{2\cos 2p}$. Equation of normal: $y-k\sin 2p=\dfrac{\sin p}{2\cos 2p}(x-k\cos p)$
2(a)(ii)	From **(a)(i)**, when $t=p=0$, equation of normal at $(k\cos 0, k\sin 2(0))$, i.e. $(k,0)$ is $y-k\sin 2(0)=\dfrac{\sin 0}{2\cos 2(0)}(x-k\cos 0)$, i.e. $y=0$. Hence, equation of tangent at $(k,0)$ is $x=k$.

2(b)(i)

$$A=\left(2|x|\right)\left(2|y|\right)=4\sqrt{x^2}\sqrt{y^2}=4\sqrt{x^2y^2}$$

$$=4\sqrt{36\left(1-\frac{y^2}{9}\right)y^2}$$

$$=4\sqrt{\left(36-4y^2\right)y^2}$$

$$=8\sqrt{9y^2-y^4}\ \text{(shown)}$$

2(b)(ii)

At stationary points,

$$\frac{\mathrm{d}A}{\mathrm{d}y}=\frac{8}{2}\left(9y^2-y^4\right)^{-\frac{1}{2}}\left(18y-4y^3\right)=\frac{8y\left(9-2y^2\right)}{\sqrt{9y^2-y^4}}=0$$

$$\Rightarrow 8y\left(9-2y^2\right)=0$$

$$\Rightarrow y=0\text{ (rejected as }y\neq 0\text{) or }y=\pm\sqrt{\frac{9}{2}}=\pm\frac{3}{\sqrt{2}}=\pm\frac{3}{2}\sqrt{2}.$$

Method 1: First derivative test

y	$\left(\pm\frac{3}{2}\sqrt{2}\right)^-$	$\pm\frac{3}{2}\sqrt{2}$	$\left(\pm\frac{3}{2}\sqrt{2}\right)^+$
$\frac{\mathrm{d}A}{\mathrm{d}y}$	+ve	0	-ve
Slope			

Method 2: Second derivative test

$$\sqrt{9y^2-y^4}\,\frac{\mathrm{d}A}{\mathrm{d}y}=8(9y-2y^3)$$

$$\sqrt{9y^2-y^4}\,\frac{\mathrm{d}^2A}{\mathrm{d}y^2}+\frac{\mathrm{d}A}{\mathrm{d}y}\left[\frac{\mathrm{d}}{\mathrm{d}y}\left(\sqrt{9y^2-y^4}\right)\right]=8(9-6y^2)$$

When $y=\pm\frac{3}{2}\sqrt{2}$, $4.5\frac{\mathrm{d}^2A}{\mathrm{d}y^2}+0=8(-18)\Rightarrow\frac{\mathrm{d}^2A}{\mathrm{d}y^2}=-32<0.$

Hence, A is a maximum when $y=\pm\frac{3}{2}\sqrt{2}$.

Required area $=8\sqrt{9\left(\pm\frac{3}{2}\sqrt{2}\right)^2-\left(\pm\frac{3}{2}\sqrt{2}\right)^4}$

$=8\sqrt{9\left(\frac{9}{2}\right)-\frac{81}{4}}=8\sqrt{\frac{81}{4}}$

$=36$ units^2.

3(a)

$a-\mathrm{i}$ is a root of $z^3+4(1+\mathrm{i})z^2+(-2+9\mathrm{i})z-5+\mathrm{i}=0$

$$\Rightarrow(a-\mathrm{i})^3+4(1+\mathrm{i})(a-\mathrm{i})^2+(-2+9\mathrm{i})(a-\mathrm{i})-5+\mathrm{i}=0$$

$$\Rightarrow(a^3-3a^2\mathrm{i}-3a+\mathrm{i})+4(1+\mathrm{i})(a^2-2a\mathrm{i}-1)\ -2a+4+(3+9a)\mathrm{i}=0$$

	$\Rightarrow a^3-5a+4-3a^2\text{i}+4\text{i}+9a\text{i}+4(a^2-2a\text{i}-1+a^2\text{i}+2a-\text{i})=0$ $\Rightarrow (a^3+4a^2+3a)+(a^2+a)\text{i}=0$. (shown) Comparing real and imaginary parts, $a(a+1)(a+3)=0 \Rightarrow a=0$ or $a=-1$ or $a=-3$ (N.A) $a(a+1)=0 \Rightarrow a=0$ or $a=-1$. Let the third root be w. Then $z^3+4(1+\text{i})z^2+(-2+9\text{i})z-5+\text{i}$ $=[z-(-\text{i})][z-(-1-\text{i})][z-(w)]=(z+\text{i})(z+1+\text{i})(z-w)$ Comparing constant term, $\text{i}(1+\text{i})(-w)=-5+\text{i}$ $\Rightarrow w=\dfrac{-5+\text{i}}{-\text{i}(1+\text{i})}=\dfrac{-5+\text{i}}{1-\text{i}}\times\dfrac{1+\text{i}}{1+\text{i}}=\dfrac{-5+\text{i}-5\text{i}-1}{2}=-3-2\text{i}.$ Hence, all the roots are $-\text{i}$, $-1-\text{i}$ and $-3-2\text{i}$. $z^3+8(1+\text{i})z^2+4(-2+9\text{i})z+8(-5+\text{i})=0$ $\dfrac{1}{8}z^3+(1+\text{i})z^2+\dfrac{1}{2}(-2+9\text{i})z+(-5+\text{i})=0$ $\left(\dfrac{1}{2}z\right)^3+4(1+\text{i})\left(\dfrac{1}{2}z\right)^2+(-2+9\text{i})\left(\dfrac{1}{2}z\right)+(-5+\text{i})=0$ $\Rightarrow \dfrac{1}{2}z=-\text{i}$ or $-1-\text{i}$ or $-3-2\text{i}$ (from first part) $\Rightarrow z=-2\text{i}$ or $-2-2\text{i}$ or $-6-4\text{i}$
3(b) (i)	$\left\|\left(1-\sqrt{3}\text{i}\right)^4\right\|=\left\|1-\sqrt{3}\text{i}\right\|^4=\left(\sqrt{1^2+(-\sqrt{3})^2}\right)^4=2^4=16.$ $\arg\left(1-\sqrt{3}\text{i}\right)^4=4\arg\left(1-\sqrt{3}\text{i}\right)=4\left(-\dfrac{\pi}{3}\right)=-\dfrac{4}{3}\pi.$ $\therefore \arg\left(1-\sqrt{3}\text{i}\right)^4=-\dfrac{4\pi}{3}+2\pi=\dfrac{2}{3}\pi.$ (for principal argument)
3(b) (ii)	$\dfrac{w^n}{w^*}$ is real $\Rightarrow \arg\left(\dfrac{w^n}{w^*}\right)=k\pi$, where $k\in\mathbf{Z}$ $\Rightarrow \arg\left(w^n\right)-\arg\left(w^*\right)=k\pi$ $\Rightarrow n\arg\left(w\right)+\arg\left(w\right)=k\pi$ $\Rightarrow \dfrac{2\pi}{3}(n+1)=k\pi$ (from first part) OR $n=\dfrac{3}{2}k-1$ Hence, the required values of n are 2, 5 and 8. **Alternatively**, $\dfrac{w^n}{w^*}$ is real $\Rightarrow \text{Im}\left(\dfrac{w^n}{w^*}\right)=0 \Rightarrow \sin\left[\arg\left(\dfrac{w^n}{w^*}\right)\right]=0$ $\Rightarrow \sin\left[\dfrac{2\pi}{3}(n+1)\right]=0 \Rightarrow \dfrac{2\pi}{3}(n+1)=k\pi$, where $k\in\mathbf{Z}$. Hence, the required values of n are 2, 5 and 8.

4(i)	$\mathbf{a} \perp \mathbf{b} - \mathbf{a} \Rightarrow \mathbf{a}\cdot(\mathbf{b}-\mathbf{a}) = 0$ $\Rightarrow \mathbf{a}\cdot\mathbf{b} = \mathbf{a}\cdot\mathbf{a} = \lvert\mathbf{a}\rvert^2 = 1$ (since $\lvert\mathbf{a}\rvert = 1$) $\Rightarrow \lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert\cos\theta = 1$ where θ is the angle between $\mathbf{a}$ and $\mathbf{b}$ $\Rightarrow \cos\theta = \dfrac{1}{\lvert\mathbf{b}\rvert} < 1 \Rightarrow \lvert\mathbf{b}\rvert > 1$. (shown) $\lvert\mathbf{b}\rvert = 2 \Rightarrow \cos\theta = \dfrac{1}{2} \Rightarrow \theta = 60^{\circ}$.
4(ii) (a)	Let α be the required angle. $\cos\alpha = 0.6 \Rightarrow \alpha = \cos^{-1} 0.6 = 53.1^{\circ}$. (1 d.p.)
4(ii) (b)	$\lambda = \cos 90^{\circ} = 0$. $0.6^2 + 0^2 + \mu^2 = 1 \Rightarrow \mu^2 = 1 - 0.36 = 0.64 \Rightarrow \mu = \pm 0.8$.
4(iii)	By Ratio Theorem, $\mathbf{b} = \dfrac{\mathbf{a}+3\mathbf{c}}{4} \Rightarrow \mathbf{c} = \dfrac{4\mathbf{b}-\mathbf{a}}{3}$. $\lvert\mathbf{c}\cdot\mathbf{a}\rvert = \left\lvert\left(\dfrac{4\mathbf{b}-\mathbf{a}}{3}\right)\cdot\mathbf{a}\right\rvert = \left\lvert\dfrac{4}{3}(\mathbf{b}\cdot\mathbf{a}) - \dfrac{1}{3}(\mathbf{a}\cdot\mathbf{a})\right\rvert$ $= \left\lvert\dfrac{4}{3}(\mathbf{a}\cdot\mathbf{a}) - \dfrac{1}{3}(\mathbf{a}\cdot\mathbf{a})\right\rvert$ (from (**i**)) $= \lvert\mathbf{a}\cdot\mathbf{a}\rvert = \left\lvert\lvert\mathbf{a}\rvert^2\right\rvert = \lvert\mathbf{a}\rvert^2 = \mathbf{a}\cdot\mathbf{a}$. (shown) Required length = $\dfrac{\lvert\mathbf{c}\cdot\mathbf{a}\rvert}{\lvert\mathbf{a}\rvert} = \dfrac{\mathbf{a}\cdot\mathbf{a}}{\lvert\mathbf{a}\rvert} = \dfrac{\lvert\mathbf{a}\rvert^2}{\lvert\mathbf{a}\rvert} = \lvert\mathbf{a}\rvert$.
4(iv)	$\lvert\mathbf{c}\times\mathbf{a}\rvert$ = Twice the area of triangle OAC $\lvert\mathbf{b}\times\mathbf{a}\rvert$ = Twice the area of triangle $OAB = (AB)h$, where h is the height of triangle OAB with AB as the base. Since A, B and C are collinear, $\lvert\mathbf{c}\times\mathbf{a}\rvert = (AC)h$. Hence, $\dfrac{\lvert\mathbf{c}\times\mathbf{a}\rvert}{\lvert\mathbf{b}\times\mathbf{a}\rvert} = \dfrac{(AC)h}{(AB)h} = \dfrac{AC}{AB} = \dfrac{4}{3}$.

Section B

5(i)	Mean score = $\dfrac{1}{5} + \dfrac{2}{6} + \dfrac{3}{7} + \dfrac{4}{8} + 5p + 6q = 3.5 \Rightarrow 5p + 6q = \dfrac{214}{105}$ … (1) $\dfrac{1}{5} + \dfrac{1}{6} + \dfrac{1}{7} + \dfrac{1}{8} + p + q = 1 \Rightarrow p + q = \dfrac{307}{840}$ … (2) From graphing calculator, $p = \dfrac{13}{84}$ and $q = \dfrac{59}{280}$.
5(ii)	Let X be the score of the die and Y be the outcome of coin. For $w = 0, 1$, $\mathrm{P}(W = w) = \mathrm{P}(X = w+1,\ Y = \text{tail}) = \dfrac{1}{2}\mathrm{P}(X = w+1)$. For $w = 2, 3, 4, 5$,

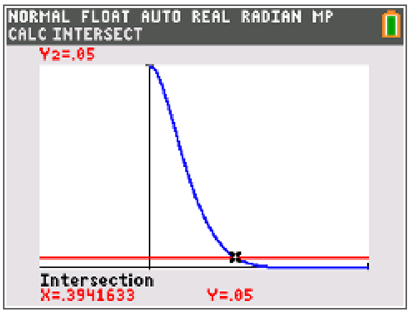
	$\mathrm{P}(W=w)=\mathrm{P}(X=w+1,\ Y=\text{tail})+\mathrm{P}(X=w-1,\ Y=\text{head})$ $=\frac{1}{2}\left[\mathrm{P}(X=w-1)+\mathrm{P}(X=w+1)\right]$ For w = 6, 7, $\mathrm{P}(W=w)=\mathrm{P}(X=w-1,\ Y=\text{head})=\frac{1}{2}\mathrm{P}(X=w-1)$. The probability distribution of W is given by: <table><tr><td>w</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td></tr><tr><td>$\mathrm{P}(W=w)$</td><td>$\frac{1}{10}$</td><td>$\frac{1}{12}$</td><td>$\frac{6}{35}$</td><td>$\frac{7}{48}$</td><td>$\frac{25}{168}$</td><td>$\frac{47}{280}$</td><td>$\frac{13}{168}$</td><td>$\frac{59}{560}$</td></tr></table>
5(iii)	Since the mean score of the die is 3.5 and the coin is fair, the expected winnings is \$3.5, hence I would not play this game since amount pay = \$4 > \$3.5 = expected winnings.

6(i)	Number of ways = $8!\times{}^{9}\mathrm{C}_2\times 2!\times 2!=5806080$. OR Number of ways = $10!\times 2!-9!\times 2!\times 2=5806080$.
6(ii)	Number of ways = $(5-1)!\times 3!\times 4!\times 3!\times 12=248832$.
6(iii)	Case 1: All 4 letters are distinct Number of code-words = ${}^{4}\mathrm{C}_4\times 4!=24$. Case 2: Exactly 3 distinct letters Number of code-words = $\left(\frac{4!}{2!}\right)\times(6)=12\times 6=72$. OR $\left({}^{4}\mathrm{C}_2\times 2!\right)\times\left({}^{2}\mathrm{C}_1\times{}^{3}\mathrm{C}_2\right)=12\times 6=72$. OR $\left({}^{4}\mathrm{C}_2\times 2!\times 2\times 2\right)+\left({}^{4}\mathrm{C}_2\times 2!\times 2\right)=48+24=72$. Case 3: Exactly 2 distinct letters Number of code-words = $\left({}^{4}\mathrm{C}_1\times 3\right)+\left({}^{4}\mathrm{C}_2\right)=12+6=18$. Total number of code-words = 24 + 72 + 18 = 114.

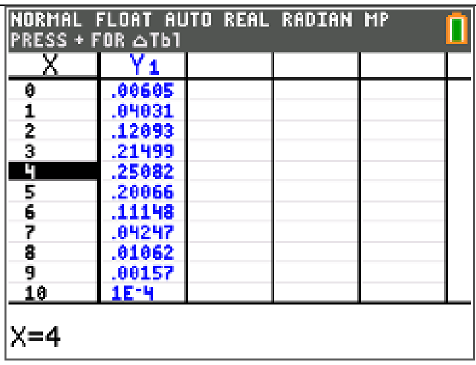
7(i)	$\mathrm{P}(B)=\frac{130}{400}=\frac{13}{40}$ and $\mathrm{P}(C)=\frac{150}{400}=\frac{3}{8}$. Since $\mathrm{P}(B\cap C)=\frac{70}{400}=\frac{7}{40}\neq\frac{39}{320}=\frac{13}{40}\times\frac{3}{8}=\mathrm{P}(B)\mathrm{P}(C)$, B and C are not independent.
7(ii)(a)	$\mathrm{P}\left(B\cup C'\right)=\frac{320}{400}=\frac{4}{5}$.
7(ii)(b)	$\mathrm{P}\left(C\mid M\cap B\right)=\frac{10}{30}=\frac{1}{3}$.
7	Let x be the number of students studying Further Mathematics out of the 400 students and F be the event that the chosen student is studying Further Mathematics. $\mathrm{P}(F\cap C)=\mathrm{P}(F)\ \mathrm{P}(C)$ (since F and C are independent) $=\frac{3}{8}\times\frac{x}{400}=\frac{3x}{3200}$

	Since $F \subseteq M, F \cap C \subseteq M \cap C$. Hence, $\mathrm{P}(F) \le \mathrm{P}(M)$ and $\mathrm{P}(F \cap C) \le \mathrm{P}(M \cap C)$ $\Rightarrow \mathrm{P}(F) \le \dfrac{250}{400} = \dfrac{5}{8}$ and $\mathrm{P}(F \cap C) \le \dfrac{40}{400} = \dfrac{1}{10}$ $\Rightarrow \dfrac{x}{400} \le \dfrac{5}{8}$ and $\dfrac{3x}{3200} \le \dfrac{1}{10}$ $\Rightarrow x \le 250$ and $x \le \dfrac{320}{3} \Rightarrow x \le 106\dfrac{2}{3}$. Hence, required number of students is 106.

8(i)	Given: $X \sim \mathrm{N}\left(\mu, 2^2\right)$. $\mathrm{P}\left(X \le 1\right) = \mathrm{P}\left(Z \le \dfrac{1-\mu}{2}\right) = 0.1587$, where $Z \sim \mathrm{N}(0,1)$ $\Rightarrow \dfrac{1-\mu}{2} \approx -0.9998 \Rightarrow \mu = 3$. (nearest integer)
8(ii)	By symmetry, $\dfrac{(3k)+(9k+4)}{2} = \mu = 3 \Rightarrow k = \dfrac{1}{6}$.
8(iii)	$\mathrm{E}(\bar{Y}) = \mathrm{E}(Y) = 10 - \mathrm{E}(X) = 10 - 3 = 7$. $\mathrm{Var}(\bar{Y}) = \dfrac{\mathrm{Var}(Y)}{2} = \dfrac{\mathrm{Var}(X)}{2} = \dfrac{4}{2} = 2$. $\mathrm{P}\left(\bar{Y} > 6\right) \approx 0.76025 = 0.760$. (3 s.f.)

9(i)	Two assumptions are: 1. The probability of a student being male is the same for **each** student. 2. Whether a student is male is independent of other students.
9(ii)	$\mathrm{P}(X \le 1) = 0.05 \Rightarrow (1-p)^{10} + 10p(1-p)^9 = 0.05$ $\Rightarrow (1-p)^9(1+9p) = 0.05$  From graphing calculator, $p = 0.394$. (3 s.f.)
9(iii)	Let Y be the number of samples of 10 students, out of 8, with at least 2 male students. Then $Y \sim \mathrm{B}(8, \mathrm{P}(X \ge 2))$, i.e. $Y \sim \mathrm{B}(8, 0.95)$. Required probability $= \mathrm{P}(Y = 7) \approx 0.27933 = 0.279$. (3 s.f.) OR ${}^8\mathrm{C}_7(0.95)^7(0.05) = 0.279$. (3 s.f.)

9(iv)



x	$P(X=x)$
3	0.21499
4	0.25082
5	0.20066

From graphing calculator, the highest probability occurs when $X = 4$, so most probable number of male students in a sample of 10 is 4, i.e. most probable number of female students = 10 − 4 = 6.

9(v)

Let S be the sum of 60 independent observations of X.
$E(S) = 60E(X) = 60(10)(0.4) = 240$.
$Var(S) = 60Var(X) = 60(10)(0.4)(0.6) = 144$.
Since $n = 60$ is large, by Central Limit Theorem,
$S \sim N(240, 144)$ approximately.
$P(S > 230) \approx 0.79767 = 0.798$. (3 s.f.)

10(i)

Unbiased estimate of population mean mass,

$$\bar{x} = \frac{\sum(x-600)}{50} + 600 = \frac{-8}{50} + 600 = 599.84.$$

Unbiased estimate of population variance,

$$s^2 = \frac{1}{50-1}\left[\sum(x-600)^2 - \frac{1}{50}\left(\sum(x-600)\right)^2\right]$$

$$= \frac{1}{49}\left[11.3 - \frac{1}{50}(-8)^2\right] = \frac{10.02}{49} = \frac{501}{2450} = 0.204. \text{ (3 s.f.)}$$

10(ii)

Let X be the mass, in grams, of a randomly chosen packet of cereal.
$H_0 : \mu = 600$ (distributor's claim)
$H_1 : \mu < 600$ (claim is overstated)
Under H_0, since $n = 50$ is large, by Central Limit Theorem,

$$\bar{X} \sim N\left(600, \frac{1}{50} \times \frac{501}{2450}\right), \text{i.e. } N\left(600, \frac{5.01}{1225}\right) \text{ approximately.}$$

Use a z-test at $\alpha = 0.01$.
From graphing calculator, p-value = 0.00618 (3 s.f.)
Since p-value = 0.00618 < 0.01 = α, we reject H_0.
There is sufficient evidence at 1% level of significance to conclude that the distributor has overstated the claim.

10(iii)

No. Since sample size <u>$n = 50$ is large</u>, <u>by Central Limit Theorem</u>, the sample mean mass of a packet of cereal will be <u>approximately normal</u>.

10(iv)

A p-value of <u>0.00618</u> means that there is a probability of <u>0.00618</u> of obtaining <u>a sample mean mass of 599.84 grams **or less**</u>, when the <u>population mean mass of cereals per packet is assumed to be 600 grams</u>.

Let Y be the mass, in grams, of a randomly chosen packet of cereal, after the change in packaging process.

$H_0 : \mu = \mu_0$ (distributor's claim)

$H_1 : \mu \neq \mu_0$ (retailer's suspicion)

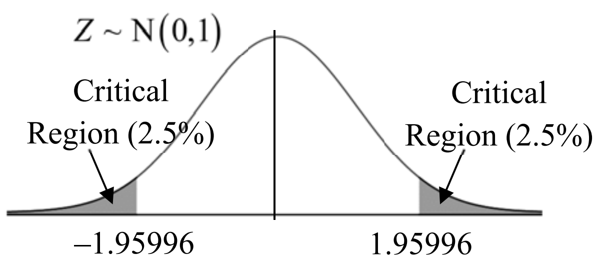
Under H_0, $\overline{Y} \sim N(\mu_0, \frac{0.5^2}{25})$, i.e. $N(\mu_0, 0.01)$.

Standardising the test statistic, $Z = \frac{\overline{Y}-\mu_0}{\sqrt{\frac{0.5^2}{25}}} \sim N(0,1)$.

Corresponding test statistic value: $z = \frac{600.6-\mu_0}{\sqrt{\frac{0.5^2}{25}}} = \frac{600.6-\mu_0}{\sqrt{0.01}}$

Critical values: –1.9599 and 1.9599

Critical region: $z \leq -1.95996$ or $z \geq 1.95996$

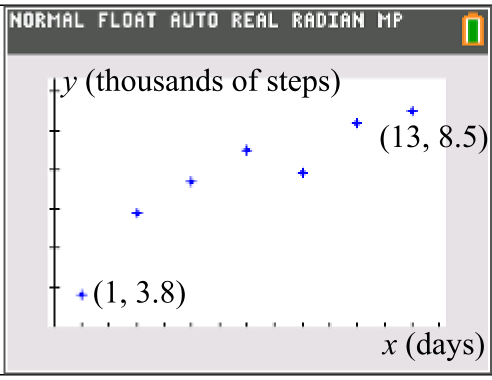


Since the retailer's suspicion is valid at 5% significance level, we **reject** H_0, $z = \frac{600.6-\mu_0}{\sqrt{\frac{0.5^2}{25}}}$ **falls in** the critical region.

$\Rightarrow \frac{600.6-\mu_0}{\sqrt{0.01}} \leq -1.95996$ or $\frac{600.6-\mu_0}{\sqrt{0.01}} \geq 1.95996$

$\Rightarrow 600.7960 \leq \mu_0$ or $600.4040 \geq \mu_0$

$\Rightarrow \mu_0 \leq 600.40$ or $\mu_0 \geq 600.80$ (2 d.p.)

11(i)	
11(ii)	A possible reason is that the step tracker is not working properly on Day 9.
11(iii)	y on x line: $y = 0.34969x + 4.4354$ (5 s.f.), i.e. $y = 0.350x + 4.44$ (3 s.f.). x on y line: $x = 2.4769y - 10.094$ (5 s.f.), i.e. $x = 2.48y - 10.1$ (3 s.f.).
11(iv)	350 is the **expected** increase in the number of steps when the time increases by 1 day. OR Kenny is expected to increase 350 steps every day.

11(v)	Since x is the independent variable and y is the dependent variable, we should use the y on x regression line. When $y = 10$, $10 = 0.34969x + 4.4354 \Rightarrow x = \dfrac{10-4.4354}{0.34969} \approx 15.913 = 16$. (nearest whole number) The estimated number of days = 16.
11(vi)	For $L = 10.1$, $r = -0.98533$ (5 d.p.) The most appropriate value for L is 10.1 since its corresponding value of $\lvert r \rvert$ is closest to 1 as compared to that of 10.2 and 10.3.
11 (vii)	The required equation is $\ln(10.1 - y) = 1.8184 - 0.10859x$ (5 s.f.) Hence, $a = 1.82$. (3 s.f.) and $b = -0.109$. (3 s.f.). When $x = 10$, $\ln(10.1 - y) = 1.8184 - 0.10859(10)$ $\Rightarrow y = 10.1 - e^{1.8184-1.0859} \approx 8.01973 = 8020$. (3 s.f.) Kenny is expected to clock 8020 steps on Day 10. This estimate is reliable since $\lvert r \rvert = 0.98533$ is close to 1 and $x = 10$ is within the range of the data set.